

CUSTOMER BEHAVIOUR ANALYSIS

End-to-End Data Analytics Project

Tools: Python • MySQL • Power BI

1. Project Overview

This project analyzes customer shopping behaviour using transactional data from around 5000 customer purchases across multiple product categories. The objective was to identify meaningful patterns in customer spending, purchasing preferences, subscription trends, and revenue generation through data analysis and visualization, helping support more informed and strategic business decisions.

5,000	30	3.62	\$994K	\$198.84
Customer Records	Unique Items	Avg. Review	Total Spend	Avg. Spend

2. Dataset Summary

Attribute	Details
Type	Retail / E-commerce Dataset
Nature	Structured (Tabular)
Level	Customer Transaction Level
Rows	5,051 (5,050 records + header)
Columns	17

3. Domain Background & Its Significance

In the e-commerce industry, businesses closely monitor customer interactions to better understand consumer behavior and purchasing trends. This includes analyzing:

- Products purchased by customers
- Purchase frequency and shopping habits
- Customer spending patterns
- Responses to discounts and promotional offers

Analyzing customer data enables businesses to:

- Identify high-value and loyal customers
- Understand customer purchase behavior and preferences
- Develop effective pricing and discount strategies
- Improve the efficiency of marketing campaigns
- Enhance customer retention and overall customer lifetime value (CLV)

4. Business Problem Statement

How can a retail e-commerce company better understand customer purchasing behaviour to increase revenue, improve customer retention, and optimize marketing and discount strategies?

Business Challenge	Key Questions
Low Customer Retention	Who are frequent buyers? Who are one-time buyers?
Ineffective Discounts	Do discounts actually increase purchase value or reduce margins?
Product Demand Gaps	Which category and products generate the highest revenue?
Payment Behaviour	Which payment methods are preferred? Are digital payments growing?
Seasonal Demand	Does season impact sales? Which season drives maximum purchases?
No Customer Segmentation	How do we identify high-value, frequent, and discount-sensitive customers?

5.Data Dictionary

Column Name	Data Type	Description
Customer ID	Numeric	Unique identifier for each customer
Age	Numeric	Age of the customer
Gender	Categorical	Male / Female / Other
Item Purchased	Categorical	Specific product purchased
Category	Categorical	Product category (Clothing, Electronics, Accessories, Footwear, Outerwear)
Purchase Amount (USD)	Numeric	Amount spent by the customer
Location	Categorical	Customer's US state

Size	Categorical	Product size (S, M, L, XL)
Color	Categorical	Product color
Season	Categorical	Season of purchase (Spring/Summer/Fall/Winter)
Review Rating	Numeric	Customer rating for the product (1–5)
Subscription Status	Categorical	Whether customer is subscribed (Yes/No)
Shipping Type	Categorical	Shipping method selected
Discount Applied	Categorical	Whether a discount was applied (Yes/No)
Previous Purchases	Numeric	Number of past purchases by the customer
Payment Method	Categorical	Mode of payment (Cash, Card, PayPal, etc.)
Frequency of Purchases	Categorical	Purchase frequency (Weekly, Monthly, etc.)

6. Exploratory Data Analysis (Python)

Data preparation and cleaning was performed in Python using pandas before passing the cleaned dataset to SQL for business analysis.

Data Loading:

- Imported pandas
- Loaded the dataset using:

```
df=pd.read_csv("customer_shopping_behavior.csv")
```
- Performed initial exploration with `df.head()`, `df.info()`, `df.describe(include='all')`, and `df.isnull().sum()` to understand structure, data types, and missing values.

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	Weekly
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card	Quarterly
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash	Monthly
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store pickup	No	25.0	Cash	Annually
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store pickup	No	17.0	Credit Card	Weekly

```
df.shape
```

(5050, 17)

```
[8]: df.describe(include="all")
```

```
[8]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
count	5050.000000	5050.000000	5050	5050	5050	4494.000000	5050	4680	5050	5050	4449.000000	5050	5050	5050	4502.0000		
unique	NaN	NaN	3	30	5	NaN	54	4	25	5	NaN	2	6	2			
top	NaN	NaN	Male	Shirt	Clothing	NaN	New York	M	Black	Spring	NaN	No	Standard	No			
freq	NaN	NaN	3018	320	2130	NaN	337	1981	404	1280	NaN	3462	1213	2789			
mean	2519.570891	44.150495	NaN	NaN	NaN	144.765236	NaN	NaN	NaN	NaN	3.668195	NaN	NaN	NaN	25.221		
std	1470.402964	15.282328	NaN	NaN	NaN	275.590101	NaN	NaN	NaN	NaN	0.865357	NaN	NaN	NaN	14.521		
min	1.000000	18.000000	NaN	NaN	NaN	10.120000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	0.000		
25%	1252.250000	31.000000	NaN	NaN	NaN	41.000000	NaN	NaN	NaN	NaN	3.000000	NaN	NaN	NaN	13.000		
50%	2499.500000	44.000000	NaN	NaN	NaN	65.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	NaN	25.000		
75%	3740.750000	57.000000	NaN	NaN	NaN	89.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	38.000		
max	5099.000000	70.000000	NaN	NaN	NaN	1499.760000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	50.000		

```

]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 5050 entries, 0 to 5049
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID           5050 non-null  int64  
1   Age                   5050 non-null  int64  
2   Gender                5050 non-null  str     
3   Item Purchased        5050 non-null  str     
4   Category              5050 non-null  str     
5   Purchase Amount (USD) 4494 non-null  float64 
6   Location              5050 non-null  str     
7   Size                  4680 non-null  str     
8   Color                 5050 non-null  str     
9   Season                5050 non-null  str     
10  Review Rating         4449 non-null  float64 
11  Subscription Status   5050 non-null  str     
12  Shipping Type         5050 non-null  str     
13  Discount Applied      5050 non-null  str     
14  Previous Purchases    4502 non-null  float64 
15  Payment Method        5050 non-null  str     
16  Frequency of Purchases 5050 non-null  str     
dtypes: float64(3), int64(2), str(12)
memory usage: 670.8 KB

```

```

: df.isnull().sum()

: Customer ID           0
  Age                  0
  Gender               0
  Item Purchased       0
  Category             0
  Purchase Amount (USD) 556
  Location             0
  Size                 370
  Color                0
  Season              0
  Review Rating       601
  Subscription Status  0
  Shipping Type       0
  Discount Applied    0
  Previous Purchases  548
  Payment Method      0
  Frequency of Purchases 0
dtype: int64

```

Data Consistency Check:

- Checked unique values in Item Purchased and Category columns.
- Found inconsistencies between item names and their assigned categories.
- Created a mapping dictionary and corrected category values to ensure accurate grouping and correct business insights.

Missing Value - Review Rating: Imputed using the product-level median rating (not global average) for each item to keep ratings realistic and meaningful

Missing Value - Size:

Applied domain-specific logic:

- Electronics & Accessories → 'Not Applicable' (as they don't have specific sizes S,M,L etc.)
- Footwear → 'Free Size'
- Clothing → mode of that category.

Missing Value - Purchase Amount: Replaced with the mean value to maintain product-level pricing consistency and avoid revenue distortion.

Missing Value - Previous Purchases: Replaced with 0, as missing values imply no prior purchase history - important for customer segmentation accuracy and loyalty analysis.

Duplicate Handling: Identified and removed (50) duplicate records based on Customer ID, ensuring each customer appears only once for accurate reporting.

Column Standardization: Renamed all columns using:
for SQL compatibility and Power BI readability.

```
df.columns=df.columns.str.replace(" ", "_")
```

```
df.columns=df.columns.str.lower()
```

Database Integration: Connected Python script to MySQL using SQLAlchemy and loaded the cleaned DataFrame into the database for SQL-based analysis.

Data Export to MySQL Server

- Installed pymysql and used SQLAlchemy's create_engine to establish the database connection
- Configured connection with host, port, username, password, and target database (customer_behaviour)
- Verified successful connection before export
- Exported cleaned DataFrame to MySQL using df.to_sql() with if_exists="replace" and index=False
- Data loaded into table customer_behaviour_analysis inside the MySQL database

7. Data Analysis using SQL (Business Queries)

12 structured business questions were answered using MySQL, covering aggregations, subqueries, window functions (RANK, PARTITION BY), and CTEs.

#	Business Question	SQL Technique Used	Findings																																																								
1	Which category generates the highest revenue?	GROUP BY, SUM, ORDER BY	<table><tr><th>category</th><th>highest_revenue</th></tr><tr><td>Electronics</td><td>486366.96</td></tr><tr><td>Accessories</td><td>194866.87</td></tr><tr><td>Clothing</td><td>183188.39</td></tr><tr><td>Footwear</td><td>111236.2</td></tr><tr><td>Outerwear</td><td>18524</td></tr></table>	category	highest_revenue	Electronics	486366.96	Accessories	194866.87	Clothing	183188.39	Footwear	111236.2	Outerwear	18524																																												
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Outerwear	18524																																																										
2	Do discounts actually increase purchase value?	GROUP BY discount_applied, AVG, SUM	<table><tr><th>discount_applied</th><th>highest_revenue</th><th>avg_revenue</th></tr><tr><td>No</td><td>502241.64</td><td>182.24</td></tr><tr><td>Yes</td><td>491940.78</td><td>219.22</td></tr></table>	discount_applied	highest_revenue	avg_revenue	No	502241.64	182.24	Yes	491940.78	219.22																																															
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3	Revenue split by gender and category	GROUP BY gender + category	<table><tr><th>gender</th><th>total_revenue</th></tr><tr><td>Male</td><td>390964.53</td></tr><tr><td>Female</td><td>352060.29</td></tr><tr><td>Other</td><td>251157.6</td></tr></table> <table><tr><th>gender</th><th>category</th><th>revenue</th></tr><tr><td>Female</td><td>Accessories</td><td>56998.81</td></tr><tr><td>Female</td><td>Clothing</td><td>61327.21</td></tr><tr><td>Female</td><td>Electronics</td><td>186064.7</td></tr><tr><td>Female</td><td>Footwear</td><td>41768.58</td></tr><tr><td>Female</td><td>Outerwear</td><td>5901</td></tr></table> <table><tr><th>gender</th><th>category</th><th>revenue</th></tr><tr><td>Male</td><td>Accessories</td><td>89260.11</td></tr><tr><td>Male</td><td>Clothing</td><td>93483.68</td></tr><tr><td>Male</td><td>Electronics</td><td>144975.7</td></tr><tr><td>Male</td><td>Footwear</td><td>50622.03</td></tr><tr><td>Male</td><td>Outerwear</td><td>12623</td></tr></table> <table><tr><td>Other</td><td>Accessories</td><td>48607.95</td></tr><tr><td>Other</td><td>Clothing</td><td>28377.51</td></tr><tr><td>Other</td><td>Electronics</td><td>155326.55</td></tr><tr><td>Other</td><td>Footwear</td><td>18845.59</td></tr></table>	gender	total_revenue	Male	390964.53	Female	352060.29	Other	251157.6	gender	category	revenue	Female	Accessories	56998.81	Female	Clothing	61327.21	Female	Electronics	186064.7	Female	Footwear	41768.58	Female	Outerwear	5901	gender	category	revenue	Male	Accessories	89260.11	Male	Clothing	93483.68	Male	Electronics	144975.7	Male	Footwear	50622.03	Male	Outerwear	12623	Other	Accessories	48607.95	Other	Clothing	28377.51	Other	Electronics	155326.55	Other	Footwear	18845.59
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4	Customers using discounts but spending above average	Subquery with AVG, WHERE filter	<table><tr><th>customer_id</th><th>purchase_amount</th><th>discount_applied</th></tr><tr><td>4977</td><td>1499.76</td><td>Yes</td></tr><tr><td>4055</td><td>1493.56</td><td>Yes</td></tr><tr><td>4226</td><td>1472.22</td><td>Yes</td></tr><tr><td>4600</td><td>1460.51</td><td>Yes</td></tr><tr><td>4662</td><td>1459.59</td><td>Yes</td></tr><tr><td>4408</td><td>1456.31</td><td>Yes</td></tr><tr><td>5091</td><td>1451.07</td><td>Yes</td></tr><tr><td>4573</td><td>1441.51</td><td>Yes</td></tr><tr><td>4940</td><td>1439.75</td><td>Yes</td></tr><tr><td>4079</td><td>1438.53</td><td>Yes</td></tr></table>	customer_id	purchase_amount	discount_applied	4977	1499.76	Yes	4055	1493.56	Yes	4226	1472.22	Yes	4600	1460.51	Yes	4662	1459.59	Yes	4408	1456.31	Yes	5091	1451.07	Yes	4573	1441.51	Yes	4940	1439.75	Yes	4079	1438.53	Yes																															
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4940	1439.75	Yes																																																																	
4079	1438.53	Yes																																																																	
5	Top 5 & Bottom 5 products by avg. review rating	GROUP BY, AVG, ORDER BY, LIMIT	<table><tr><th>item_purchased</th><th>Avg_ratings</th><th>item_purchased</th><th>Avg_ratings</th></tr><tr><td>Gloves</td><td>3.86</td><td>Phone</td><td>2.94</td></tr><tr><td>Sandals</td><td>3.84</td><td>Headphones</td><td>3.08</td></tr><tr><td>Boots</td><td>3.82</td><td>Watch</td><td>3.1</td></tr><tr><td>Hat</td><td>3.8</td><td>Bag</td><td>3.17</td></tr><tr><td>Skirt</td><td>3.79</td><td>Laptop</td><td>3.19</td></tr></table>	item_purchased	Avg_ratings	item_purchased	Avg_ratings	Gloves	3.86	Phone	2.94	Sandals	3.84	Headphones	3.08	Boots	3.82	Watch	3.1	Hat	3.8	Bag	3.17	Skirt	3.79	Laptop	3.19																																								
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Hat	3.8	Bag	3.17																																																																
Skirt	3.79	Laptop	3.19																																																																
6	Average purchase: all shipping types compared	GROUP BY shipping_type, AVG	<table><tr><th>shipping_type</th><th>orders_placed</th><th>avg_purchase</th><th>revenue</th></tr><tr><td>Express</td><td>1199</td><td>358.12</td><td>429380.98</td></tr><tr><td>Standard</td><td>1201</td><td>340.57</td><td>409020.45</td></tr><tr><td>Free Shipping</td><td>675</td><td>60.41</td><td>40777</td></tr><tr><td>Store Pickup</td><td>650</td><td>59.89</td><td>38931</td></tr><tr><td>2-Day Shipping</td><td>627</td><td>60.73</td><td>38080</td></tr><tr><td>Next Day Air</td><td>648</td><td>58.63</td><td>37993</td></tr></table>	shipping_type	orders_placed	avg_purchase	revenue	Express	1199	358.12	429380.98	Standard	1201	340.57	409020.45	Free Shipping	675	60.41	40777	Store Pickup	650	59.89	38931	2-Day Shipping	627	60.73	38080	Next Day Air	648	58.63	37993																																				
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Next Day Air	648	58.63	37993																																																																
7	Subscribed vs. non-subscribed spending comparison	GROUP BY subscription_status, AVG	<table><tr><th>subscription_status</th><th>users</th><th>avg_revenue</th><th>total_revenue</th></tr><tr><td>No</td><td>3421</td><td>165.72</td><td>566937.02</td></tr><tr><td>Yes</td><td>1579</td><td>270.58</td><td>427245.4</td></tr></table>	subscription_status	users	avg_revenue	total_revenue	No	3421	165.72	566937.02	Yes	1579	270.58	427245.4																																																				
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No	3421	165.72	566937.02																																																																
Yes	1579	270.58	427245.4																																																																
8	Top 5 products with highest discount usage %	CASE WHEN, COUNT, percentage calc	<table><tr><th>item_purchased</th><th>total_number_of_times_sold</th><th>number_of_times_sold_when_discount_applied</th><th>discount_percent</th></tr><tr><td>Laptop</td><td>143</td><td>74</td><td>51.75</td></tr><tr><td>Phone</td><td>171</td><td>88</td><td>51.46</td></tr><tr><td>Headphones</td><td>166</td><td>85</td><td>51.20</td></tr><tr><td>Hat</td><td>154</td><td>77</td><td>50.00</td></tr><tr><td>Watch</td><td>157</td><td>78</td><td>49.68</td></tr></table>	item_purchased	total_number_of_times_sold	number_of_times_sold_when_discount_applied	discount_percent	Laptop	143	74	51.75	Phone	171	88	51.46	Headphones	166	85	51.20	Hat	154	77	50.00	Watch	157	78	49.68																																								
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9	Customer segmentation: New / Returning / Loyal	CASE WHEN on previous_purchases	<table><tr><th>customer_segment</th><th>customer_count</th></tr><tr><td>Loyal Customer</td><td>3085</td></tr><tr><td>Returning Customer</td><td>1354</td></tr><tr><td>New Customer</td><td>561</td></tr></table>	customer_segment	customer_count	Loyal Customer	3085	Returning Customer	1354	New Customer	561																																																								
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New Customer	561																																																																		
10	Top 3 most purchased products per category	CTE + RANK() window function	<table><tr><th>category</th><th>item_purchased</th><th>most_purchased</th><th>rnk</th></tr><tr><td>Accessories</td><td>Jewelry</td><td>171</td><td>1</td></tr><tr><td>Accessories</td><td>Bag</td><td>164</td><td>2</td></tr><tr><td>Accessories</td><td>Sunglasses</td><td>161</td><td>3</td></tr><tr><td>Accessories</td><td>Belt</td><td>161</td><td>3</td></tr><tr><td>Clothing</td><td>Shirt</td><td>315</td><td>1</td></tr><tr><td>Clothing</td><td>Pants</td><td>171</td><td>2</td></tr><tr><td>Clothing</td><td>Blouse</td><td>171</td><td>2</td></tr><tr><td>Electronics</td><td>Phone</td><td>171</td><td>1</td></tr><tr><td>Electronics</td><td>Headphones</td><td>166</td><td>2</td></tr><tr><td>Electronics</td><td>Watch</td><td>157</td><td>3</td></tr><tr><td>Footwear</td><td>Shoes</td><td>303</td><td>1</td></tr><tr><td>Footwear</td><td>Sandals</td><td>160</td><td>2</td></tr><tr><td>Footwear</td><td>Sneakers</td><td>145</td><td>3</td></tr><tr><td>Outerwear</td><td>Jacket</td><td>163</td><td>1</td></tr><tr><td>Outerwear</td><td>Coat</td><td>161</td><td>2</td></tr></table>	category	item_purchased	most_purchased	rnk	Accessories	Jewelry	171	1	Accessories	Bag	164	2	Accessories	Sunglasses	161	3	Accessories	Belt	161	3	Clothing	Shirt	315	1	Clothing	Pants	171	2	Clothing	Blouse	171	2	Electronics	Phone	171	1	Electronics	Headphones	166	2	Electronics	Watch	157	3	Footwear	Shoes	303	1	Footwear	Sandals	160	2	Footwear	Sneakers	145	3	Outerwear	Jacket	163	1	Outerwear	Coat	161	2
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Outerwear	Jacket	163	1																																																																
Outerwear	Coat	161	2																																																																

11	Repeat buyers (>5 purchases) vs. subscription rate	CASE WHEN + PARTITION BY window	<table><tr><th>customer_type</th><th>subscription_status</th><th>customer_count</th><th>percents</th></tr><tr><td>Normal Buyers</td><td>No</td><td>658</td><td>62.85</td></tr><tr><td>Normal Buyers</td><td>Yes</td><td>389</td><td>37.15</td></tr><tr><td>Repeat Buyers</td><td>No</td><td>2763</td><td>69.90</td></tr><tr><td>Repeat Buyers</td><td>Yes</td><td>1190</td><td>30.10</td></tr></table>	customer_type	subscription_status	customer_count	percents	Normal Buyers	No	658	62.85	Normal Buyers	Yes	389	37.15	Repeat Buyers	No	2763	69.90	Repeat Buyers	Yes	1190	30.10
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12	Revenue contribution by age group	CASE WHEN age bins, GROUP BY	<table><tr><th>age_group</th><th>total_revenue</th></tr><tr><td>51+</td><td>399345.32</td></tr><tr><td>35-50</td><td>266706.7</td></tr><tr><td>26-35</td><td>177421.93</td></tr><tr><td>18-25</td><td>150708.48</td></tr></table>	age_group	total_revenue	51+	399345.32	35-50	266706.7	26-35	177421.93	18-25	150708.48										
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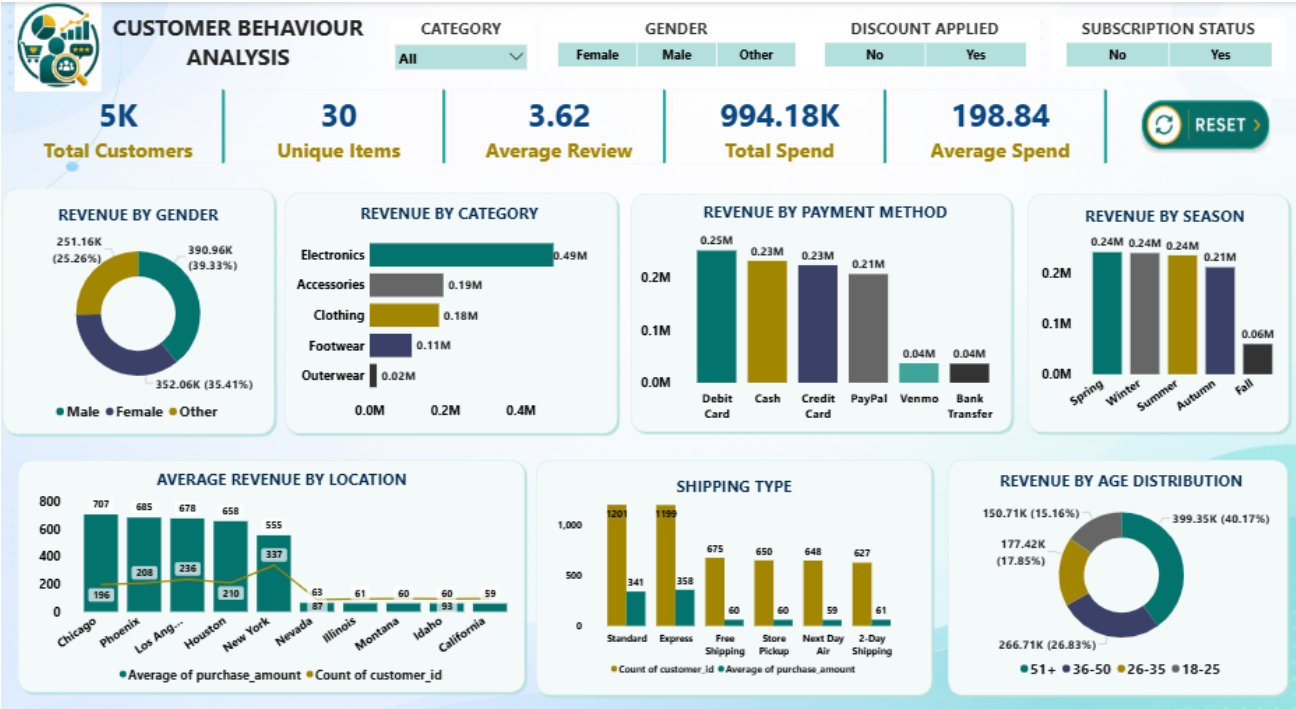
Key SQL Findings

- Electronics is the highest-revenue category at \$486K - nearly 2.5× the second-ranked Accessories (\$194K)
- Customers using discounts show a higher average purchase (\$219) vs. non-discount customers (\$182), suggesting discounts attract high-intent buyers
- Male customers generate 39.3% of total revenue vs. 35.4% Female and 25.3% Other
- Express shipping customers have a slightly higher avg. spend (\$60.48) vs. Standard (\$58.46)
- Subscribed customers (1,579) generate significantly higher avg. spend (\$270.58) vs. non-subscribers (\$165.72)
- Loyalty dominates: 3,085 Loyal customers vs. 1,354 Returning and 561 New
- Laptop, Phone, and Headphones show the highest discount dependency (>50% of sales with discount)
- 51+ age group is the top revenue contributor (\$399K), followed by 35- 50 (\$266K)

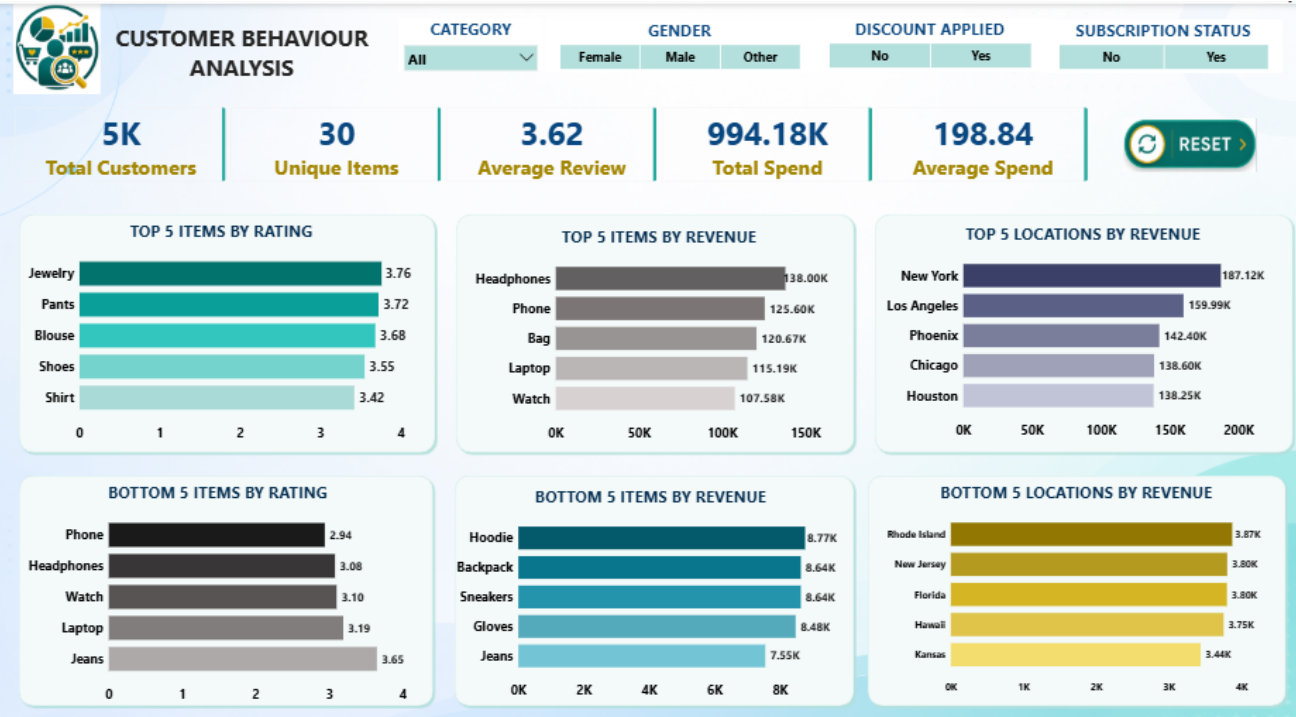
8. Power BI Dashboard

An interactive two-page Power BI dashboard was built, providing a clean and analytical interface with slicers for Category, Gender, Discount Applied, and Subscription Status and a reset button.

Page 1 : Customer Overview



Page 2 : Rankings & Performance



9. Business Recommendations

Area	Recommendation	Expected Impact
Subscription Program	Promote exclusive subscriber benefits (early access, free shipping, loyalty points)	Increase subscription rate and CLV
Discount Strategy	Retain discounts for Electronics (high dependency) but reduce on Clothing/Footwear where dependency is lower	Protect margins without losing volume
Customer Retention	Introduce a loyalty tier system: Returning - > Loyal transition incentives for the 1,354 returning customers	Reduce churn, increase repeat purchase rate
Product Positioning	Promote top-rated items (Gloves, Sandals, Boots) in marketing campaigns; invest in improving Phone and Headphone ratings	Boost conversion from high-visibility products
Shipping Upsell	Offer Express and 2-Day shipping promotions - these customers already spend more on average	Increase average order value
Age-Group Marketing	Focus campaigns on 51+ and 35–50 segments (top revenue contributors); use digital channels for 18–35 growth	Maximize ROI on marketing spend
Payment Adoption	Incentivize digital payments (PayPal, Venmo) which currently lag behind Cash and Debit Card	Reduce payment friction, increase conversion
Seasonal Planning	Align inventory and promotions with Spring/Summer peak seasons; run targeted campaigns to boost Fall revenue (\$60K vs \$240K+ other seasons)	Reduce seasonal revenue gaps

10. Conclusion

- This project delivers an end-to-end analytics workflow ,using - Python data cleaning, SQL business analysis, and Power BI visualization -across 5,050 customer records.
- **Key takeaways:** Electronics drives ~49% of revenue; subscribed customers spend significantly more; the 51+ segment is the highest-value age group; and discounts attract high-intent buyers but create dependency risk in Electronics and Accessories.
- The findings offer a data-driven foundation for smarter marketing, inventory, and retention decisions